

WEEK 1 CYBERSECURITY LAB DOCUMENTARY

VirtualBox Lab Setup: Kali Linux + Windows Server 2022

Field	Details
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Module / Track	SOC Analyst Foundation Roadmap
Week	Week 1 - Full Lab Setup
Date	19 May 2026
Lab Platform	Oracle VirtualBox

Prepared as portfolio evidence for the cybersecurity lab roadmap.

1. Objective

The objective of Week 1 was to build a safe, isolated cybersecurity lab environment using Oracle VirtualBox. The lab uses Kali Linux as the security testing machine and Windows Server 2022 as the Windows target/server machine. The main technical requirement was to make both systems communicate through the same internal network without exposing the lab to the public internet.

2. Tools and Systems Used

Component	Purpose
Oracle VirtualBox	Virtualization platform used to run the lab machines.
Kali Linux	Linux-based security testing system used for command-line verification and future SOC/security tools.
Windows Server 2022	Windows server machine used for network testing, firewall configuration, logs, and future Active Directory labs.
Internal Network	Isolated VirtualBox network used to allow VM-to-VM communication only.
Command Prompt / Terminal	Used to verify IP addresses and test connectivity.
Windows Defender Firewall with Advanced Security	Used to enable ICMPv4 Echo Request so Kali can ping Windows Server.

3. Final Lab Design

The final Week 1 lab uses two virtual machines connected to the same VirtualBox Internal Network.

Machine	Operating System	IP Address	Role
Kali VM	Kali Linux	192.168.56.10/24	Security testing / Linux verification machine
Windows Server VM	Windows Server 2022	192.168.56.20/24	Windows server / logging / future AD machine

Network name: CyberLabs

Network type: Internal Network

Subnet: 192.168.56.0/24

Default gateway: Not required for this isolated Week 1 lab

4. Setup Procedure

4.1 Kali Linux Setup

Kali Linux was installed first and configured with a static IP address of 192.168.56.10/24. The IP configuration was verified using the ip a command.

4.2 Windows Server 2022 Setup

Windows Server 2022 was installed as the second VM. It was configured with a static IP address of 192.168.56.20 and subnet mask 255.255.255.0.

4.3 VirtualBox Network Configuration

Both VMs were connected to the same VirtualBox Internal Network. The network adapter name must match exactly on both machines, otherwise the VMs are isolated from each other.

4.4 Firewall Configuration

Windows Server initially did not reply to ICMP ping requests from Kali. The File and Printer Sharing (Echo Request - ICMPv4-In) rule was enabled in Windows Defender Firewall with Advanced Security.

4.5 Connectivity Testing

Connectivity was tested in both directions: Windows Server to Kali and Kali to Windows Server. Both tests completed successfully with 0% packet loss after correcting the network name and enabling ICMPv4 on Windows Server.

5. Errors Faced and Fixes Applied

Issue	What Happened	Solution Applied
Windows Server license terms error	During the first Windows Server installation attempt, setup could not read Microsoft Software License Terms.	The VirtualBox unattended installation ISO was causing problems. The unattended ISO was removed and the Windows Server ISO was attached manually.
No bootable medium found	The VM showed that it could not read from the boot medium.	The Windows Server ISO was attached as an optical drive and the boot order was set to Optical first, Hard Disk second.
Hard disk / ISO confusion	The VM at one point had only the ISO attached, without a virtual hard disk.	A VDI virtual hard disk was added under the AHCI/SATA controller so Windows Server had a disk to install onto.
Network adapter page empty in Windows	The Windows Network Connections folder appeared empty.	The VirtualBox adapter was checked, cable connected was enabled, and Intel PRO/1000 MT Desktop adapter type was used.
Different internal network names	Kali and Windows were not communicating because one VM was on a different internal network name.	Both VMs were configured to use the same Internal Network name: CyberLabs.
Windows blocked ping from Kali	Windows could ping Kali, but Kali could not ping Windows Server.	The ICMPv4 Echo Request inbound rule was enabled in Windows Defender Firewall with Advanced Security.
Wrong command attempts	Some commands such as ip addr or ip 192.168.56.10 were typed incorrectly during testing.	The correct Linux command ip a was used to verify the Kali IP address, and ping was used for connectivity testing.

6. Evidence Screenshots

The screenshots below document the final configuration, firewall rule, network settings, and successful connectivity tests.

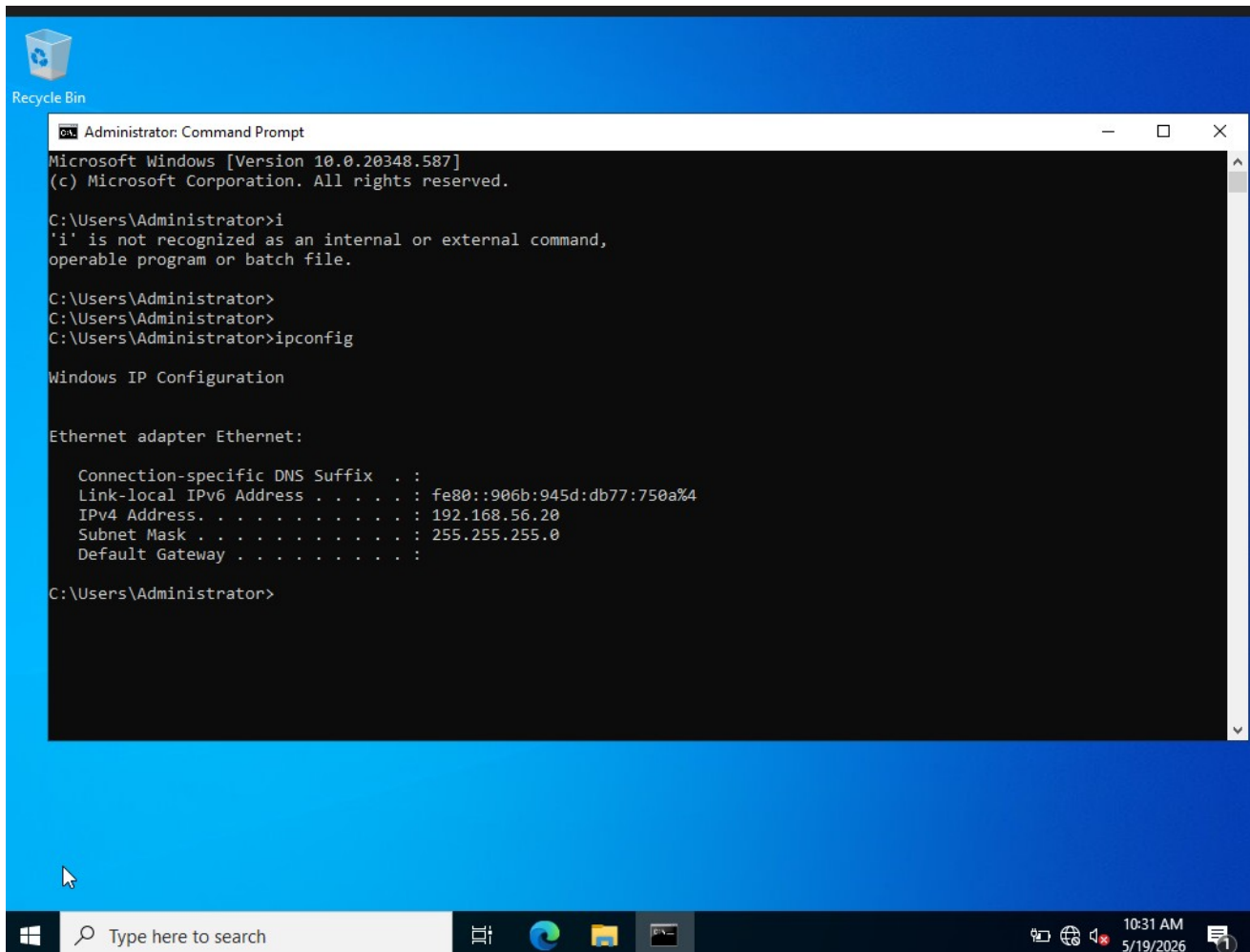


Figure 1 - Windows Server IP configuration showing IPv4 address 192.168.56.20.

```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ ip addr  
Object "addr" is unknown, try "ip help".  
  
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:c1:7b:38 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.10/24 brd 192.168.56.255 scope global noprefixroute eth0  
        valid_lft forever preferred_lft forever  
    inet6 fe80::a00:27ff:fec1:7b38/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(kali@kali)-[~]  
$
```

Figure 2 - Kali Linux successful ping test to Windows Server at 192.168.56.20.

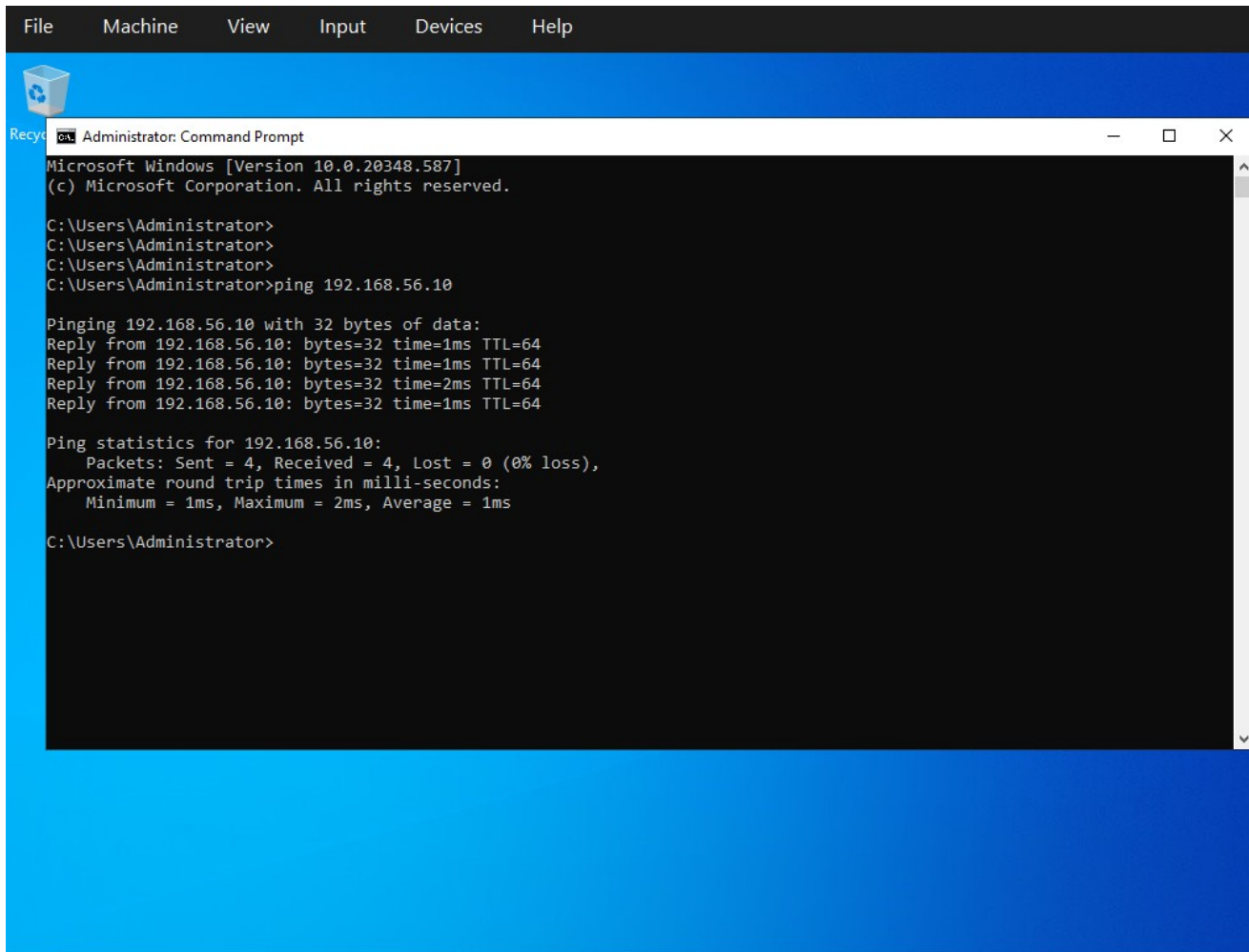


Figure 3 - Windows Server firewall overview showing the active network profile as Guest/Public.

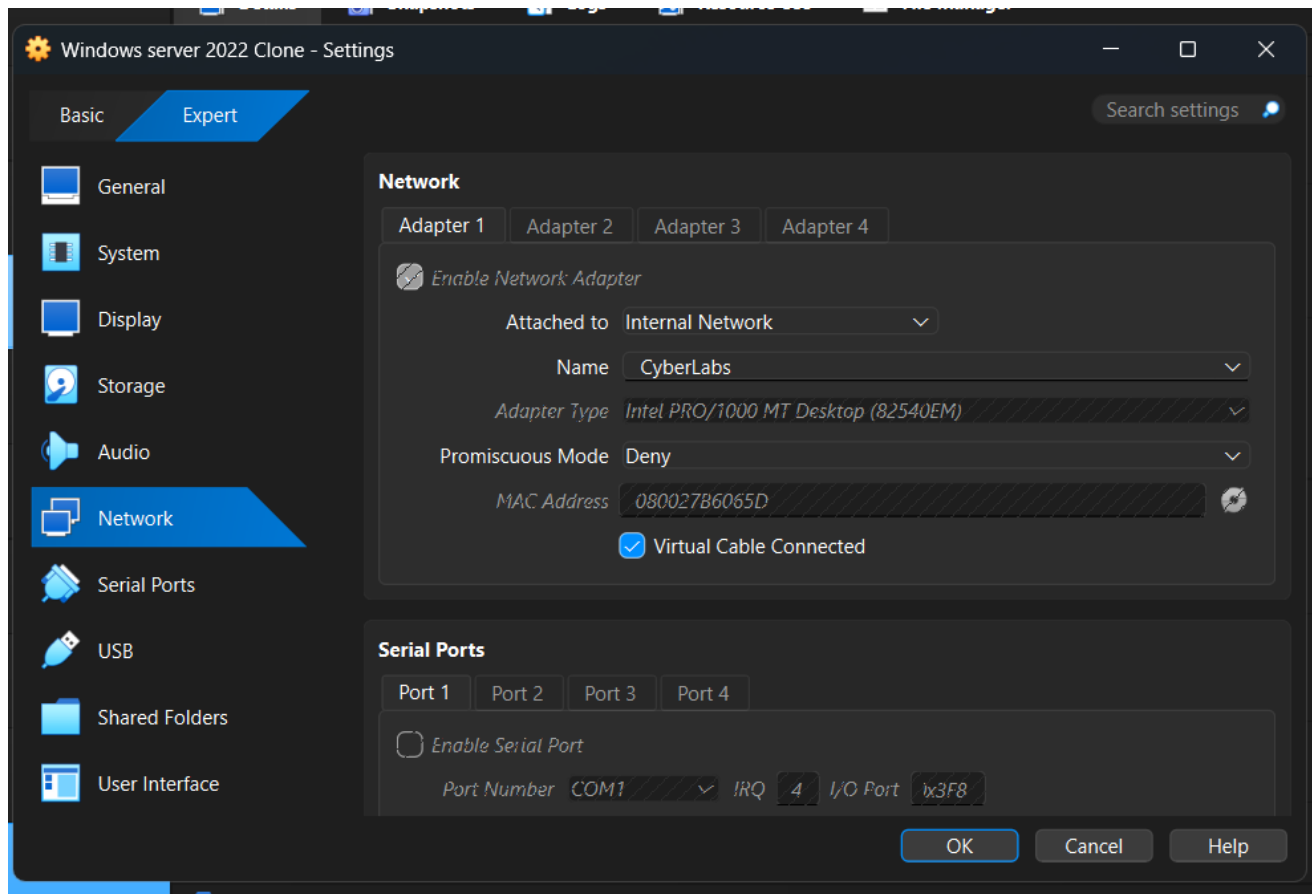


Figure 4 - Kali ping waiting before Windows firewall ICMP rule was enabled.

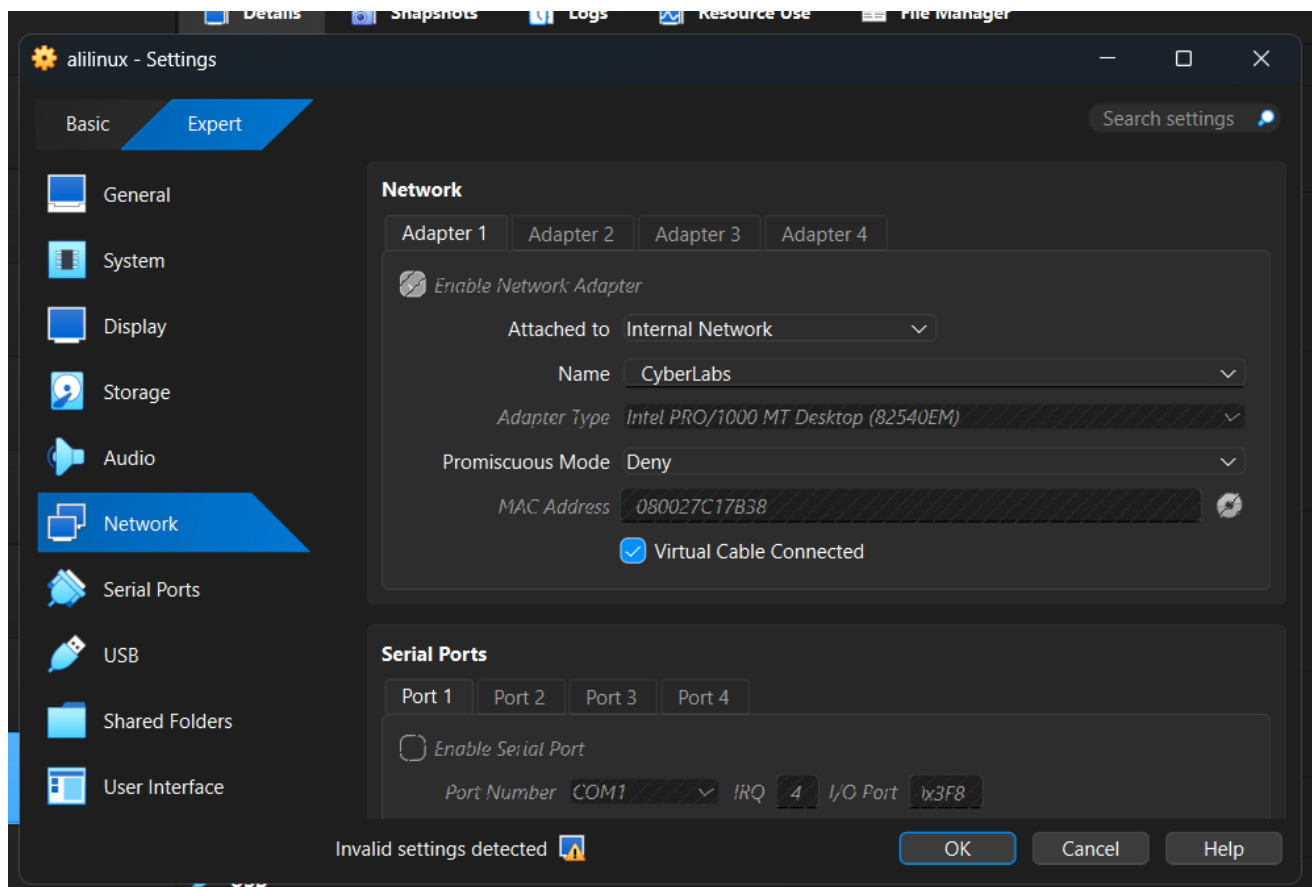


Figure 5 - Kali VirtualBox network adapter configured as Internal Network.

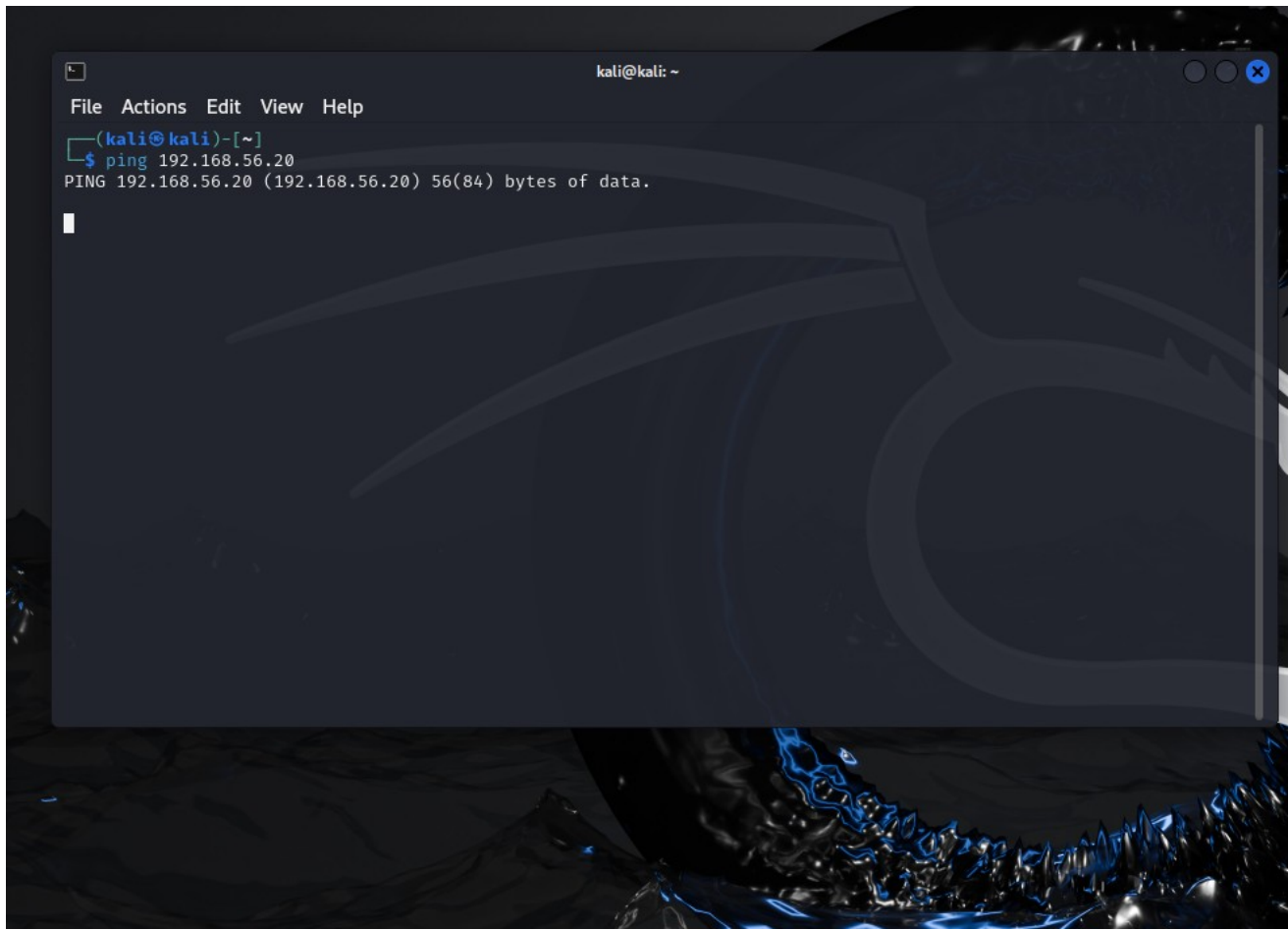


Figure 6 - Windows Server VirtualBox network adapter configured as Internal Network.

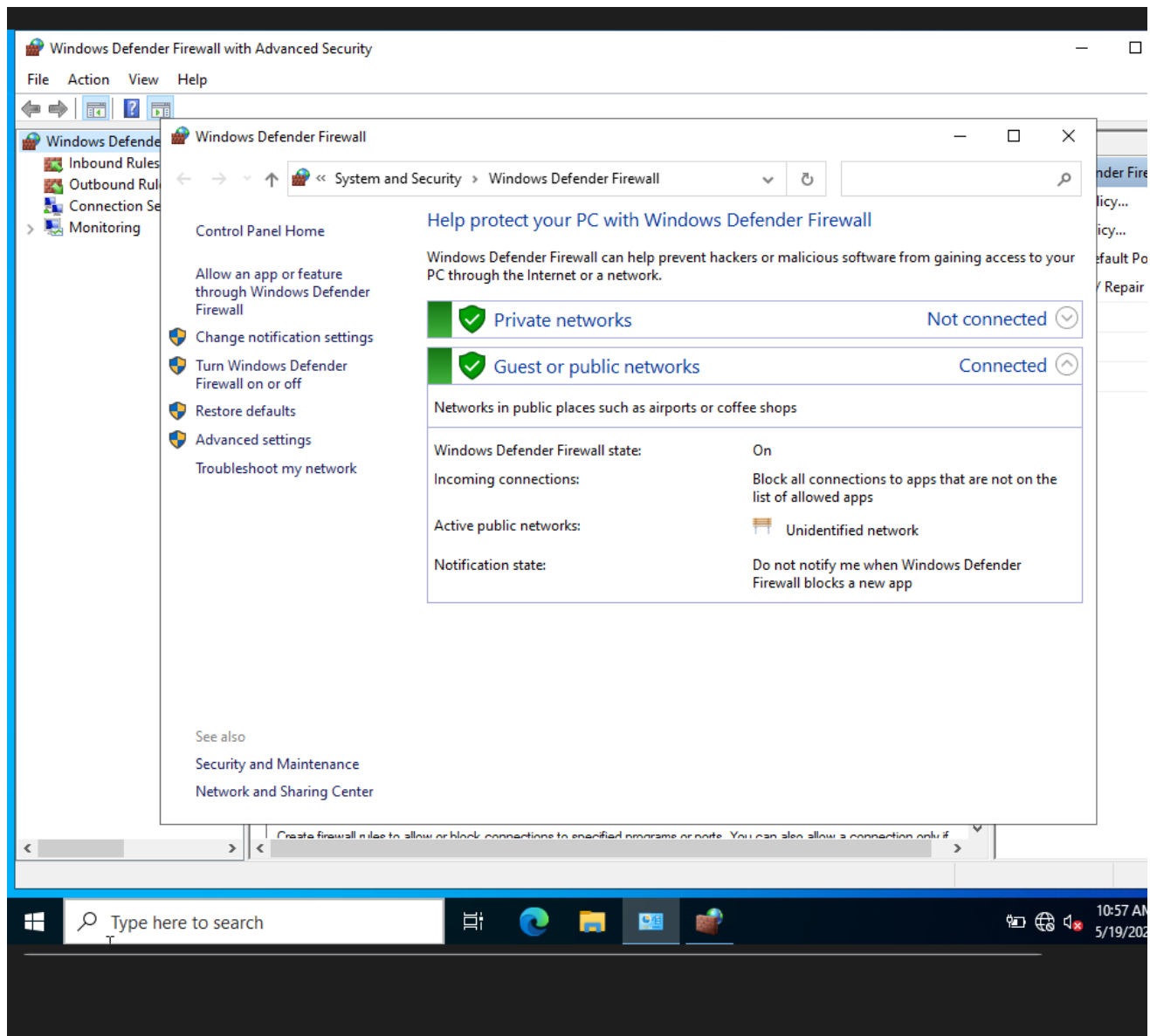


Figure 7 - Windows Server successfully pinging Kali Linux at 192.168.56.10.

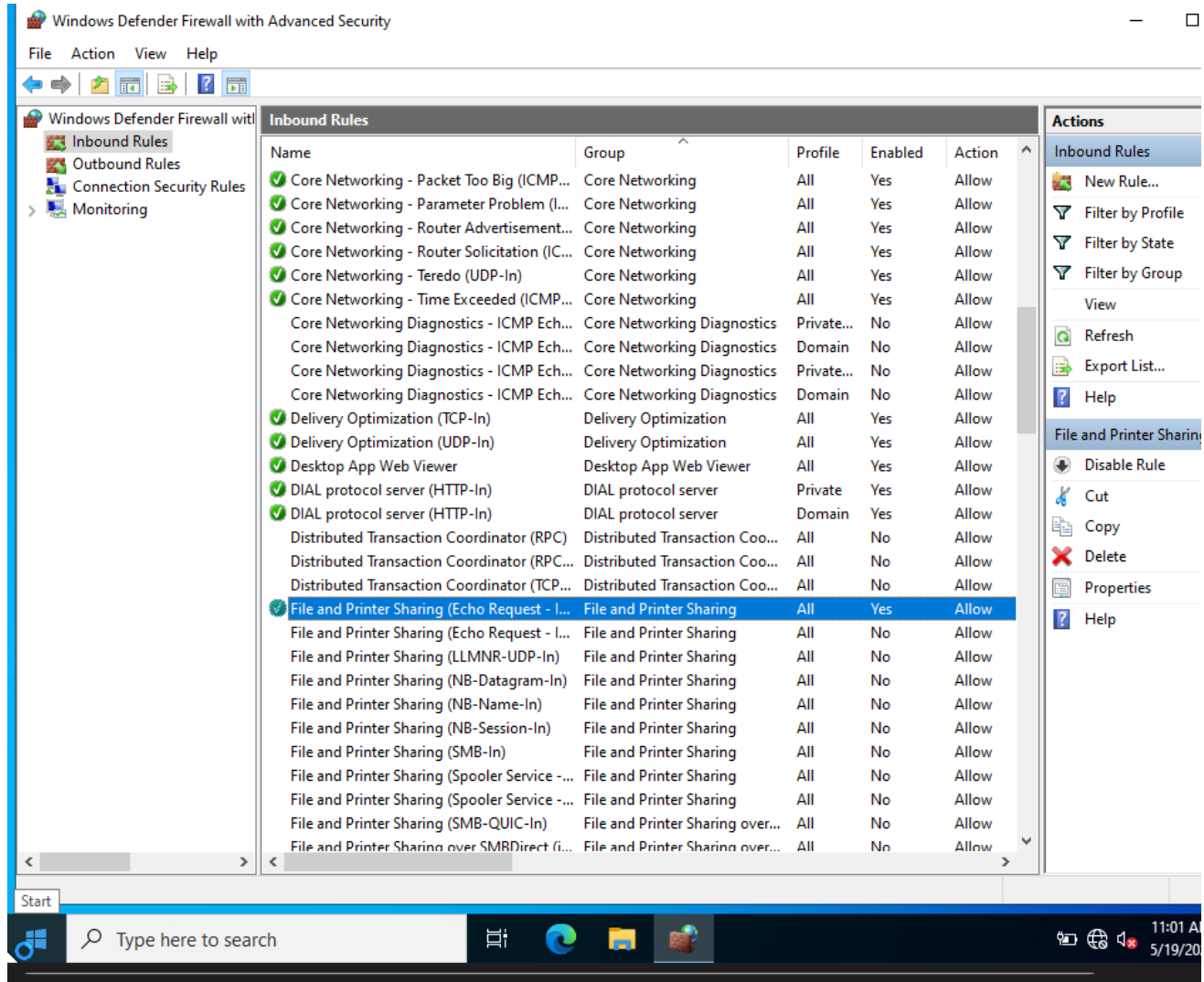


Figure 8 - Kali IP address verification using `ip a`, showing 192.168.56.10/24.

```
kali@kali: ~  
File Actions Edit View Help  
  
64 bytes from 192.168.56.20: icmp_seq=598 ttl=128 time=1.09 ms  
^C  
— 192.168.56.20 ping statistics —  
598 packets transmitted, 223 received, 62.709% packet loss, time 606365ms  
rtt min/avg/max/mdev = 0.504/1.941/24.552/2.047 ms  
  
(kali@kali)-[~]  
$ ping 192.168.56.20  
PING 192.168.56.20 (192.168.56.20) 56(84) bytes of data.  
64 bytes from 192.168.56.20: icmp_seq=1 ttl=128 time=17.6 ms  
64 bytes from 192.168.56.20: icmp_seq=2 ttl=128 time=0.632 ms  
64 bytes from 192.168.56.20: icmp_seq=3 ttl=128 time=1.85 ms  
64 bytes from 192.168.56.20: icmp_seq=4 ttl=128 time=1.23 ms  
64 bytes from 192.168.56.20: icmp_seq=5 ttl=128 time=2.47 ms  
64 bytes from 192.168.56.20: icmp_seq=6 ttl=128 time=1.55 ms  
64 bytes from 192.168.56.20: icmp_seq=7 ttl=128 time=2.00 ms  
64 bytes from 192.168.56.20: icmp_seq=8 ttl=128 time=7.89 ms  
64 bytes from 192.168.56.20: icmp_seq=9 ttl=128 time=1.20 ms  
64 bytes from 192.168.56.20: icmp_seq=10 ttl=128 time=3.59 ms  
^C  
— 192.168.56.20 ping statistics —  
10 packets transmitted, 10 received, 0% packet loss, time 9031ms  
rtt min/avg/max/mdev = 0.632/4.002/17.605/4.941 ms  
  
(kali@kali)-[~]  
$
```

Figure 9 - Windows Defender Firewall ICMPv4 Echo Request rule enabled for all profiles.

7. Final Validation

After the network name mismatch was corrected and ICMPv4 was allowed on Windows Server, both systems successfully communicated.

Test	Command	Result
Windows Server to Kali	ping 192.168.56.10	Successful: 4 sent, 4 received, 0% loss.
Kali to Windows Server	ping 192.168.56.20	Successful: 10 transmitted, 10 received, 0% packet loss.
Windows IP check	ipconfig	Windows Server confirmed as 192.168.56.20/24.
Kali IP check	ip a	Kali confirmed as 192.168.56.10/24.

8. Lessons Learned

- VirtualBox Internal Network names must match exactly for VM-to-VM communication.
- Windows Firewall can block inbound ping even when the IP configuration is correct.
- A Windows Server VM requires both a bootable ISO and a virtual hard disk for installation.
- The correct command for checking Kali IP details is ip a.
- Troubleshooting should move layer by layer: virtual adapter, IP address, firewall, then connectivity test.

9. Conclusion

Week 1 was completed successfully. A basic isolated cybersecurity lab was created in VirtualBox using Kali Linux and Windows Server 2022. Kali was configured with IP address 192.168.56.10 and Windows Server was configured with IP address 192.168.56.20. Both VMs were placed on the same Internal Network and successful ping tests confirmed two-way communication. Several realistic setup issues were encountered and solved, including Windows Server installation problems, boot media configuration, network adapter issues, mismatched VirtualBox network names, and Windows Firewall blocking ICMP requests. The lab is now ready for Week 2, which will focus on CIA, AAA, risk tables, and Windows Event Viewer.

End of Week 1 Documentary